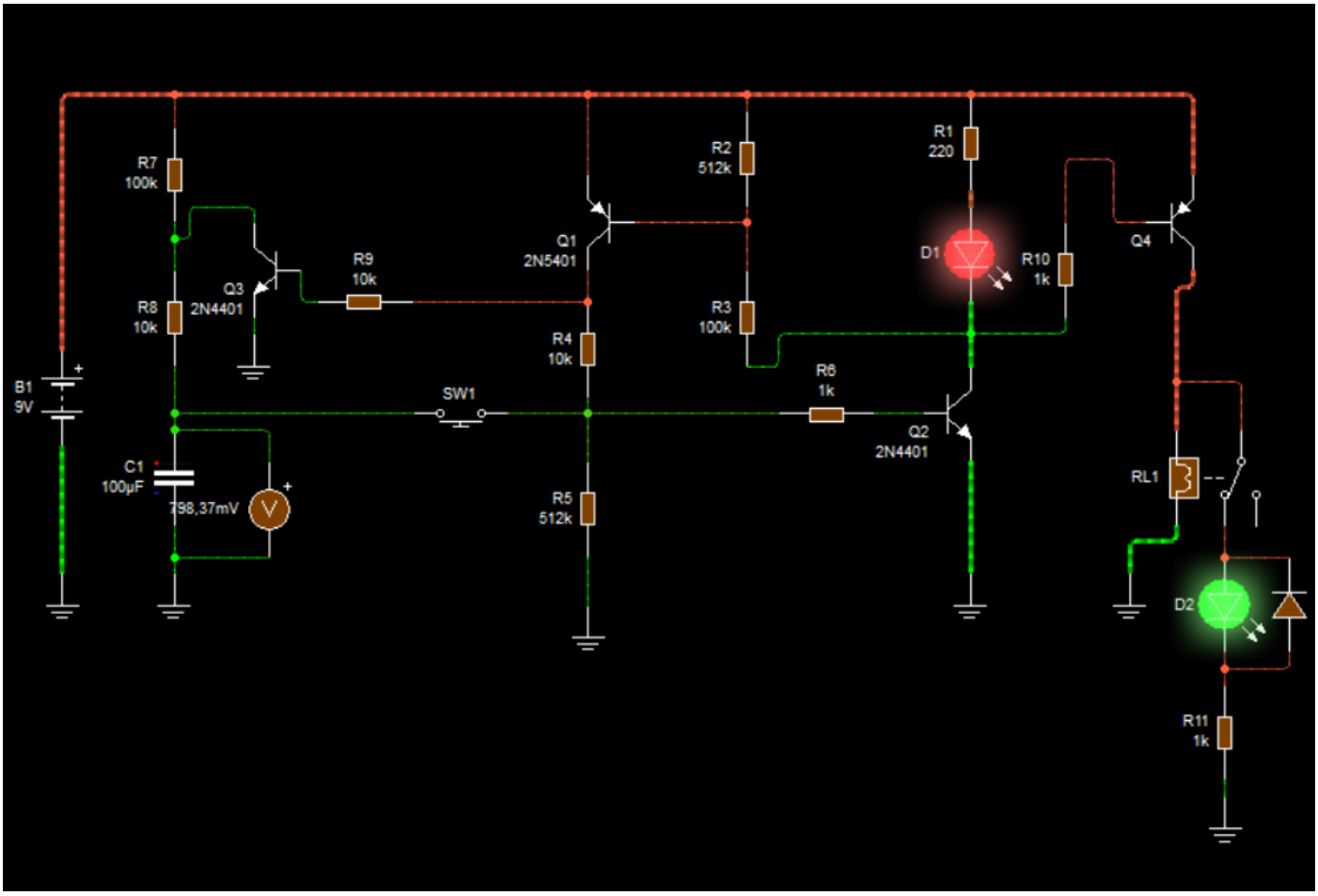
Transistor-Based Time-Delay Relay with Latching Control

This project is a 9V transistor-based timer and relay control circuit that combines a time-delay function with a bistable (memory) action. It uses only discrete components (BJTs, resistors, capacitor, diode, and relay) — no ICs.

Main Features:

- Adjustable time delay using capacitor C1 (100µF) and resistors.
- Latching (memory) behavior using a transistor feedback network.
- Relay output (RL1) to control external loads.
- Status LEDs:
- D1 (Red) → Indicates intermediate / timing state.
- D2 (Green) → Indicates relay activated.
- Flyback diode protection across the relay coil.

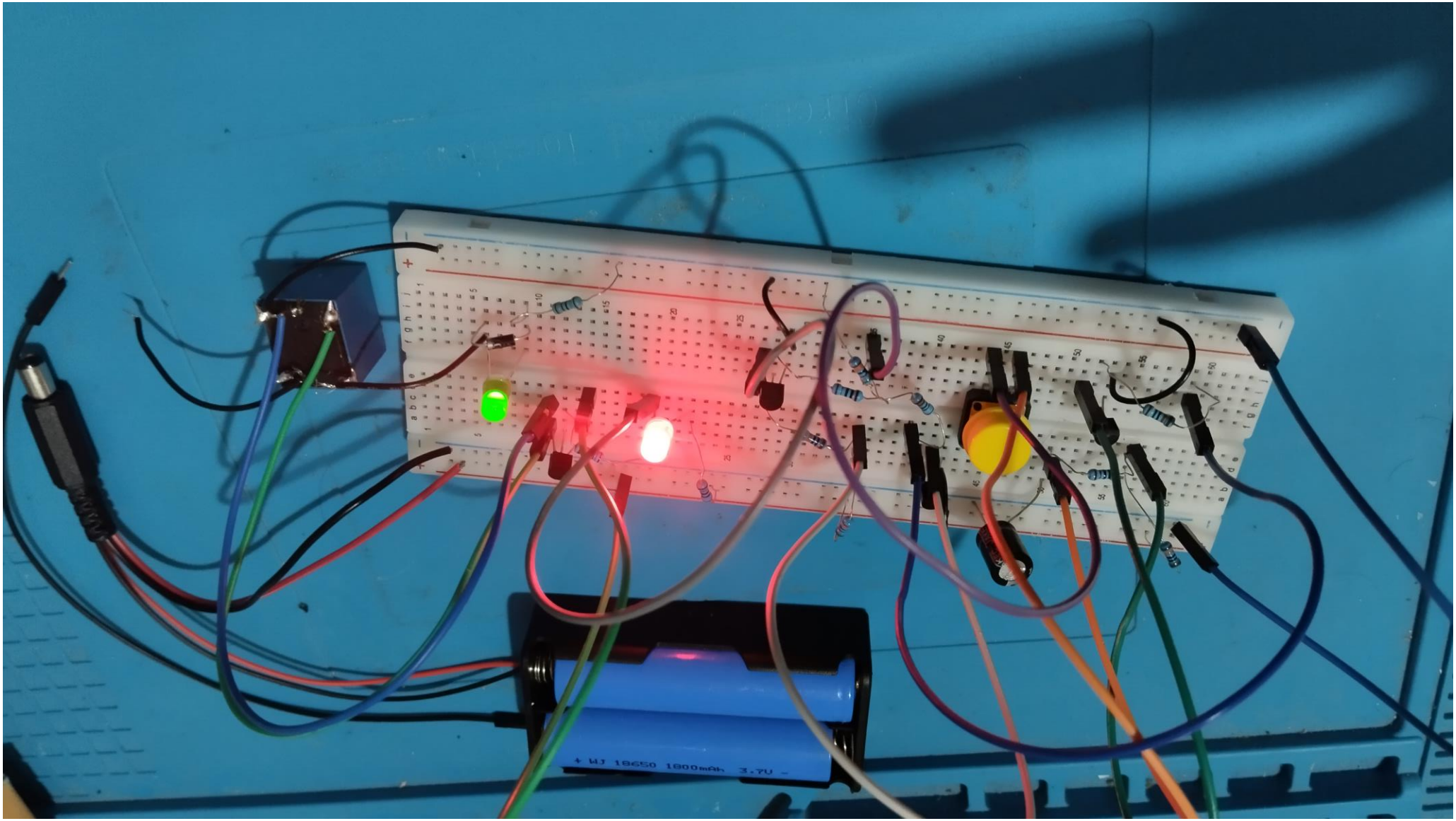


How It Works:

- Power Section (Left Side)
- The 9V battery powers the circuit.
- C1 (100µF) creates a charging delay when power is applied.
- The RC network determines how long the delay lasts.
- Control & Latch Section (Middle)
- Transistors Q1 and Q3 form a control stage that processes the delayed signal.
- The resistor network (R2, R3, R4, R5) creates a biasing and feedback system.
- SW1 allows manual triggering/resetting.
- Driver Stage (Right Side)
- Q2 acts as a switching transistor.
- When activated, it:
- Turns ON the red LED (D1).
- Drives Q4, which energizes the relay.
- The relay then powers the green LED (D2) or an external load.
- Protection
- A diode across the relay coil prevents voltage spikes when switching OFF.

Applications

- Delayed power-on relay
- Auto shutoff timer
- Soft-start control
- Simple alarm or automation system


+ WJ 18650 1800mAh 3.7V -